SOLUTIONS - Exercise - Week 4: MATH1061/7861

Problems. Prove or disprove the following statements:

(1) $\forall\, m \in \mathbb{Z}, \exists\, n \in \mathbb{Z}$, such that $m+n$ is odd and mn is even. **(3 marks)**

The statement is true and we use a direct proof to show this.

Proof. If m is even, then $m = 2k$ for some $k \in \mathbb{Z}$. Choose $n = m + 1 = 2k + 1$ which is an integer since $k \in \mathbb{Z}$.

Then $m + n = 2k + 2k + 1 = 2(2k) + 1$ and $mn = 2k(2k + 1) = 2(2k^2 + k)$. Since $k \in \mathbb{Z}$, we know that $2k \in \mathbb{Z}$ and $(2k+1) \in \mathbb{Z}$. Therefore, $m+n$ is odd and mn is even.

If m is odd, then $m = 2k + 1$ for some $k \in \mathbb{Z}$. Choose $n = m - 1 = 2k$. Then $m + n = 2(2k) + 1$ and $mn = 2(2k + 1)$ as before, and so they are odd and even respectively. □

MARKING:
1 mark for mentioning that the statement is true.
2 marks for the proof of the statement (1 mark if the proof is in the right direction, but some mistakes are made). No marks if only an example for a specific m is given.

(2) For all $x \in \mathbb{Z}$, there exists $y \in \mathbb{Z}$ such that $x = 3y + 5$. **(3 marks)**

This statement is false. If $x = 1$ then to satisfy the predicate we need $y = \frac{-4}{3}$ but that is not an integer.

MARKING:
1 mark for mentioning that the statement is false.
2 marks for a valid counter-example.

(3) There is no smallest positive real number. **(4 marks)**

This statement is true. It can be written as

$$\forall x \in \mathbb{R}_{>0},\ \exists y \in \mathbb{R}_{>0} \text{ such that } y < x$$

or equivalently as

$$\sim \left(\exists x \in \mathbb{R}_{>0} \text{ such that } \forall y \in \mathbb{R}_{>0},\ x \leq y\right).$$

To prove this statement, take any $x \in \mathbb{R}_{>0}$. Let $y = \frac{x}{2}$. Then y is a positive real number and $y < x$.

MARKING:
1 mark for interpreting the statement correctly (either by writing out a formal statement involving quantifiers, or implicitly by proceeding in the correct way for the proof).
1 mark for mentioning that the statement is false.
2 marks for the proof of the statement (1 mark if the proof is in the right direction, but some mistakes are made).